

Mode-I Stage 7: 3-Layer UEL/UMAT Architecture & State Transfer Mechanics

Layer 1: User Element U1/U3 (Phase-Field DOFs 1, 2)

- Weak Form: $(g_c/l_0 d - g_c/l_0 \nabla^2 d + 2(1-d)H) = 0$
- Writes Trial Phase Field d to Common Block: SV_PHASE_TRIAL(PHYSIDX)

Layer 2: User Element U2/U4 (Displacement DOFs 1, 2)

- Degraded Hookean Elasticity: $\sigma_{\text{mech}} = ((1-d)^2 + k)\mathbf{C}_0 : \epsilon$
- Provides TRUE System Stiffness Matrix AMATRX and Internal Force RHS = -F_INT
- Writes History H to Common Block: SV_H_TRIAL(PHYSIDX, KPT)

COMMON /CB_STATE_TRANS/
(Trial Phase d & History H Exchange)

Layer 3: Companion Facsimile CPE4/CPE3 (Standard Element / UMAT)

- Governed Package 92: STRESS(I) = 0.D0, DDSDDE(I,I) = 1.D-11 $\rightarrow$ MISESERRI $\equiv$ 0.0
- Molnár (2017) Lineage: Constitutively consistent $E_{\text{dummy}} = 10^{-11} \rightarrow \sigma \sim 10^{-12}$
- Transfers SDVs ($d, H, E_{\text{frac}}, E_{\text{elas}}$) for Standard Abaqus Field Output

SDV State Transfer
(d, H, Energy)